NKPC HSA Bayesian State-Space Estimation: Labor share gap, HP filter

July 9, 2026

1 Data and Transformations

Raw data are read from `data/raw/`. Processed model-ready data are written to `data/processed/`. The HSA competition aggregator convention is

$$N_t^{model} = \frac{100\log(N_t^{level}) - \overline{100\log(N^{level})}}{10}.$$

Thus N_t^{model}, $\hat{N}_t$, and $\bar{N}_t$ are centered and measured in ten-log-point units. If a run metadata file records another `n_transform`, interpret the corresponding coefficients using that metadata.

2 Model Specifications

The CES benchmark is

$$\pi_t = \alpha\pi_{t-1} + (1-\alpha)E_t\pi_{t+1} + \kappa x_t + e_t.$$

The dynamic HSA specification is

$$\pi_t = \alpha\pi_{t-1} + (1-\alpha)E_t\pi_{t+1} + \kappa x_t - \theta\hat{N}_t + e_t.$$

The steady HSA specification is

$$\pi_t = \alpha\pi_{t-1} + (1-\alpha)E_t\pi_{t+1} + (\kappa_0 + \delta\bar{N}_t)x_t + e_t.$$

The both-effects HSA specification is

$$\pi_t = \alpha\pi_{t-1} + (1-\alpha)E_t\pi_{t+1} + (\kappa_0 + \delta\bar{N}_t)x_t - (\theta_0 + \gamma\bar{N}_t)\hat{N}_t + e_t.$$

The maintained HSA decomposition is

$$N_t = \hat{N}_t + \bar{N}_t, \quad \hat{N}_t = \rho_1\hat{N}_{t-1} + \rho_2\hat{N}_{t-2} + u_t, \quad \bar{N}_t = n + \bar{N}_{t-1} + \epsilon_t.$$

3 Unit and Covariance Conventions

Priors for κ, κ_0, and δ are specified in physical units. Legacy Gibbs samplers internally use `KAPPA_SCALE=100` when the design matrix contains $x_t/100$. All posterior draws, tables, figures, SDDR calculations, and model-comparison summaries are reported in physical units. The default dynamic-HSA shock covariance allows only (e_t, ζ_t) correlation; diagonal and historical full covariance modes are available only through explicit run metadata. Because the default N transform is centered and divided by 10, the reported δ, θ, θ_0, and γ coefficients are already effects for a ten-log-point movement in the corresponding N component. κ_0 and θ_0 are intercepts at average $\bar{N}_t$.

4 Coefficient Summary

Table 1 reports posterior means and 95% credible intervals for key parameters across all four models (baseline prior, full sample). Cells show *mean* [*ci_2.5*, *ci_97.5*].

Table 1: Posterior means and 95% credible intervals by model

parameter	ces	hsa_steady	hsa_dynamic	hsa_full
alpha	0.640 [0.536, 0.747]	0.636 [0.529, 0.742]	0.641 [0.540, 0.744]	0.640 [0.533, 0.748]
kappa	-0.008 [-0.168, 0.150]	NaN	-0.026 [-0.181, 0.139]	NaN
kappa_0	NaN	-0.015 [-0.177, 0.145]	NaN	-0.009 [-0.179, 0.150]
delta	NaN	0.013 [-0.021, 0.047]	NaN	0.014 [-0.022, 0.048]
theta	NaN	NaN	0.035 [-0.247, 0.357]	NaN
theta_0	NaN	NaN	NaN	0.050 [-0.161, 0.291]
gamma	NaN	NaN	NaN	-0.001 [-0.038, 0.037]
rho_1	NaN	0.267 [0.056, 0.548]	0.407 [0.080, 1.117]	0.297 [0.055, 0.612]
rho_2	NaN	-0.831 [-0.969, -0.268]	-0.614 [-0.952, 0.060]	-0.833 [-0.967, -0.256]
phi_1	0.632 [0.498, 0.768]	0.633 [0.504, 0.772]	0.633 [0.503, 0.760]	0.633 [0.496, 0.764]
lambda_ez	0.010 [-0.202, 0.218]	0.019 [-0.187, 0.228]	NaN	0.017 [-0.188, 0.225]
n	NaN	-0.033 [-0.062, 0.002]	-0.030 [-0.066, 0.007]	-0.031 [-0.060, 0.002]

5 Result Blocks

Results are organized by condition. Each section fixes a data specification, competition measurement frequency, prior specification, and sample period, then reports the coefficient table, prior/posterior overlay by model, Savage-Dickey density ratios, time-varying κ_t path, and model-comparison output for that condition.

6 labor_share_gap_hp: quarterly_interpolated, prior=baseline, period=full

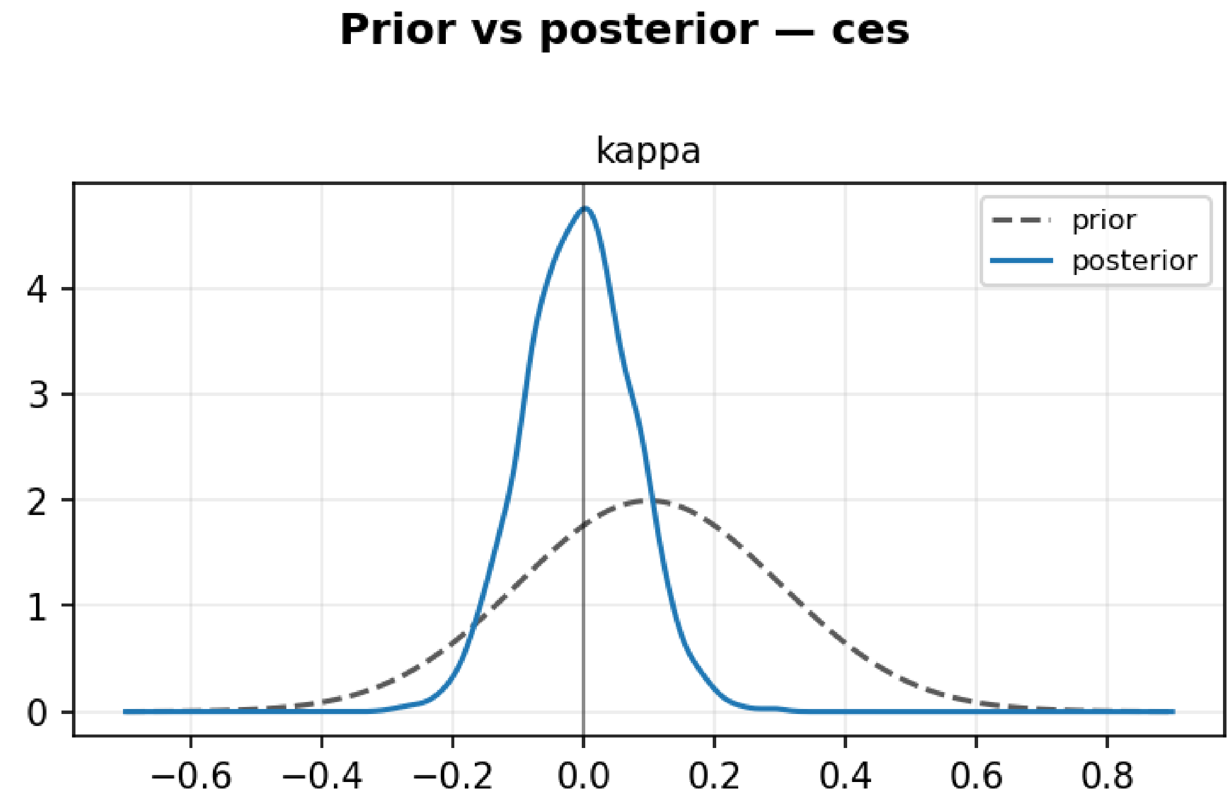
Condition. data=`labor_share_gap_hp`, competition=`quarterly_interpolated`, prior=`baseline`, period=`full`, constraint=`unrestricted` (unrestricted).

6.1 Coefficient Table

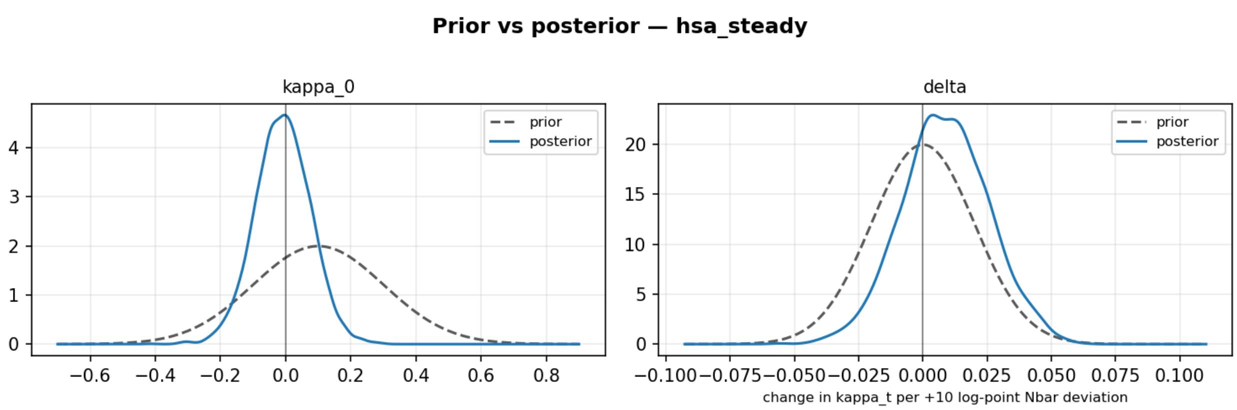
parameter	ces	hsa_steady	hsa_dynamic	hsa_full
alpha	0.640 [0.536, 0.747]	0.636 [0.530, 0.741]	0.637 [0.534, 0.737]	0.630 [0.523, 0.737]
kappa	-0.008 [-0.168, 0.150]	NaN	-0.003 [-0.164, 0.157]	NaN
kappa_0	NaN	-0.009 [-0.178, 0.157]	NaN	-0.000 [-0.160, 0.161]
delta	NaN	0.009 [-0.025, 0.042]	NaN	0.010 [-0.023, 0.043]
theta	NaN	NaN	-0.082 [-0.172, 0.026]	NaN
theta_0	NaN	NaN	NaN	-0.014 [-0.138, 0.102]
gamma	NaN	NaN	NaN	-0.027 [-0.063, 0.009]
rho_1	NaN	1.804 [1.702, 1.896]	1.767 [1.657, 1.865]	1.786 [1.672, 1.881]
rho_2	NaN	-0.817 [-0.907, -0.721]	-0.781 [-0.876, -0.676]	-0.802 [-0.894, -0.692]
phi_1	0.632 [0.498, 0.768]	0.632 [0.503, 0.768]	0.633 [0.503, 0.764]	0.632 [0.497, 0.762]
lambda_ez	0.010 [-0.202, 0.218]	0.014 [-0.199, 0.215]	NaN	0.005 [-0.200, 0.208]
n	NaN	-0.045 [-0.064, -0.021]	-0.053 [-0.071, -0.030]	-0.043 [-0.054, -0.033]

6.2 Prior vs Posterior by Model

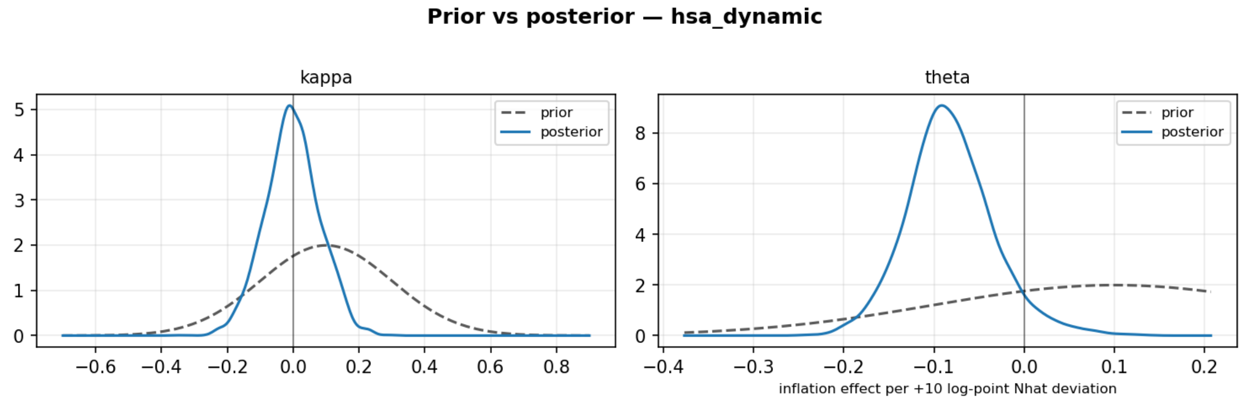
ces



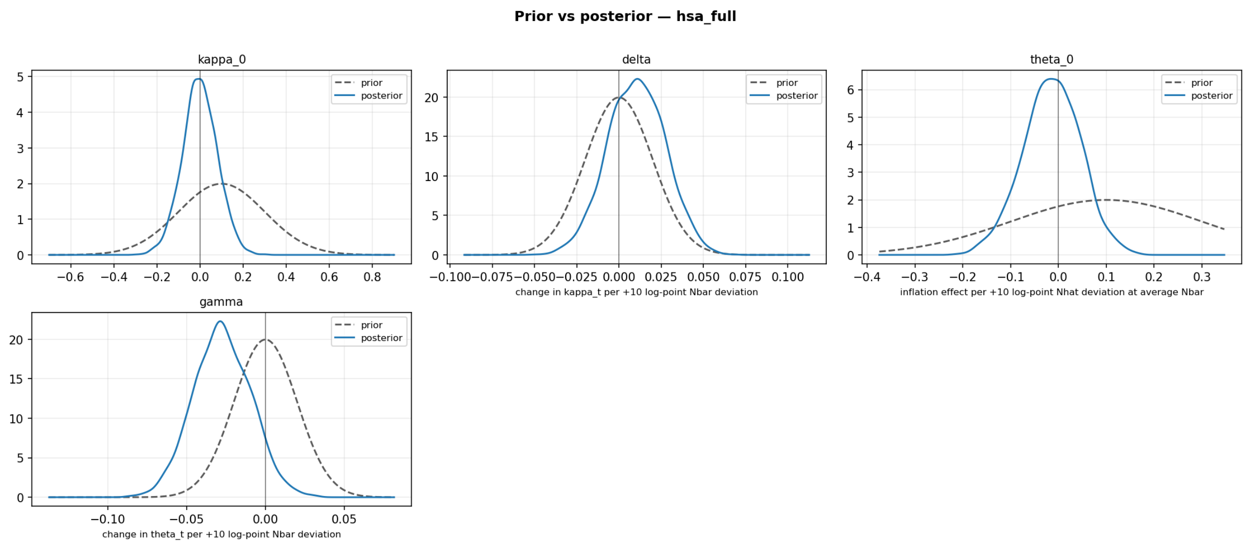
hsa_steady



hsa_dynamic


Prior vs posterior — hsa_dynamic
kappa
theta
prior
posterior
inflation effect per +10 log-point Nhat deviation

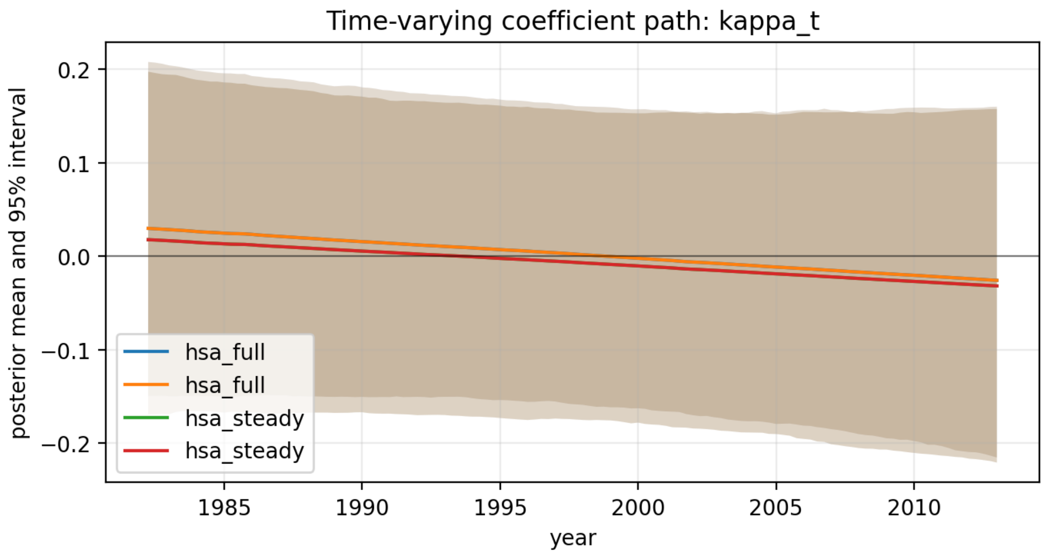
hsa_full

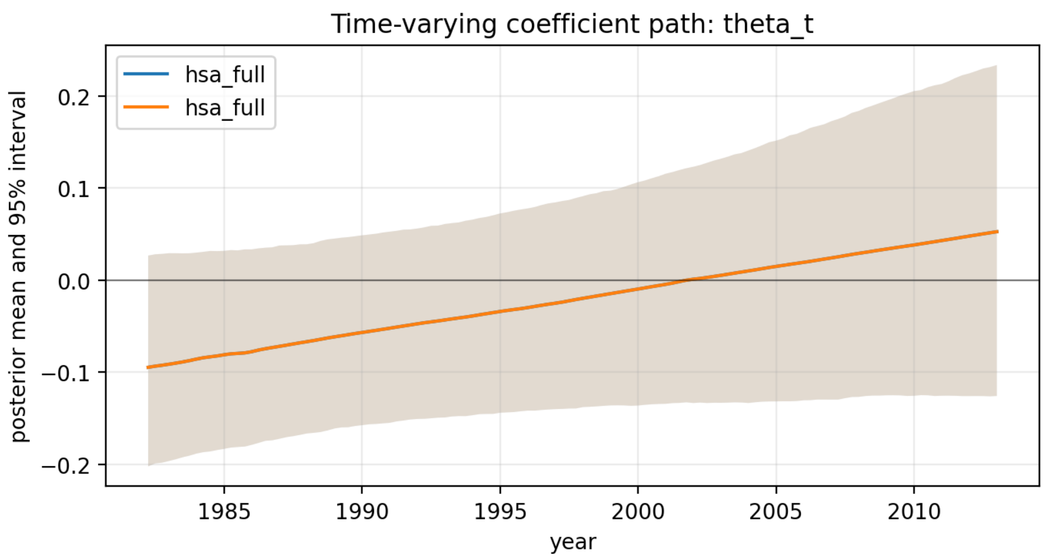

Prior vs posterior — hsa_full
kappa_0
delta
theta_0
gamma
prior
posterior
change in kappa_t per +10 log-point Nbar deviation
inflation effect per +10 log-point Nhat deviation at average Nbar
change in theta_t per +10 log-point Nbar deviation

6.3 Savage-Dickey Density Ratio

model	restriction	prior_mean	prior_sd	sddr_bf01
ces	kappa=0	0.1000	0.2000	2.7022
ces	kappa=0	0.1000	0.2000	2.7022
hsa_dynamic	kappa=0	0.1000	0.2000	2.8421
hsa_dynamic	theta=0	0.1000	0.2000	0.9151
hsa_dynamic	kappa=0	0.1000	0.2000	2.8421
hsa_dynamic	theta=0	0.1000	0.2000	0.9151
hsa_full	kappa_0=0	0.1000	0.2000	2.8031
hsa_full	delta=0	0.0000	0.0200	0.9858
hsa_full	theta_0=0	0.1000	0.2000	3.5876
hsa_full	gamma=0	0.0000	0.0200	0.3739
hsa_full	kappa_0=0	0.1000	0.2000	2.8031
hsa_full	delta=0	0.0000	0.0200	0.9858
hsa_full	theta_0=0	0.1000	0.2000	3.5876
hsa_full	gamma=0	0.0000	0.0200	0.3739
hsa_steady	kappa_0=0	0.1000	0.2000	2.6466
hsa_steady	delta=0	0.0000	0.0200	1.0765
hsa_steady	kappa_0=0	0.1000	0.2000	2.6466
hsa_steady	delta=0	0.0000	0.0200	1.0765

6.4 Time-Varying Kappa Path





6.5 Model Comparison

model	predictive_score	posterior_predictive_rmse	log_marginal_likelihood	bayes_factor_vs_baseline	sddr_delta_bf01	sddr_theta_bf01	sddr_gamma_bf01
ces	-121.9287	0.6455	-295.1854	1.0000	NaN	NaN	NaN
ces	-121.9287	0.6455	-295.1854	1.0000	NaN	NaN	NaN
hsa_dynamic	-119.1558	0.6319	-173.2199	931017367259397970905309480841633630181308971037491 20.0000	NaN	0.9151	NaN
hsa_dynamic	-119.1558	0.6319	-173.2199	93101736725939797090530948084163363018130897103749120.0000	NaN	0.9151	NaN
hsa_full	-118.4356	0.6270	-126.9439	116510469872491947495478604043981516186696575296578865598867140678223134 72.0000	0.9858	NaN	0.3739
hsa_full	-118.4356	0.6270	-126.9439	11651046987249194749547860404398151618669657529657886559886714067822313472.0000	0.9858	NaN	0.3739
hsa_steady	-121.7376	0.6442	-172.1110	28217996662962960089210181180603071896387979603397 8368.0000	1.0765	NaN	NaN
hsa_steady	-121.7376	0.6442	-172.1110	282179966629629600892101811806030718963879796033978368.0000	1.0765	NaN	NaN

7 labor_share_gap_hp: annual_q4, prior=baseline, period=full

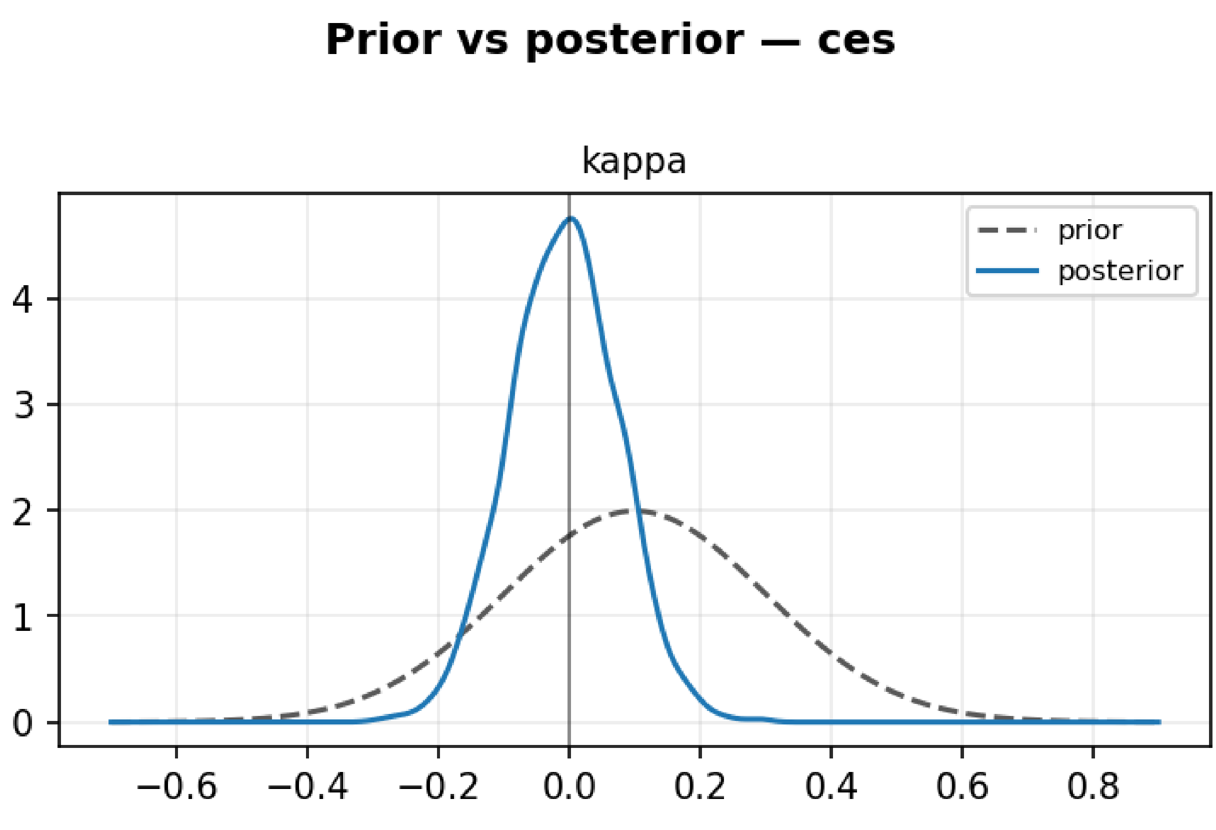
Condition. data=`labor_share_gap_hp`, competition=`annual_q4`, prior=`baseline`, period=`full`, constraint=`unrestricted` (unrestricted).

7.1 Coefficient Table

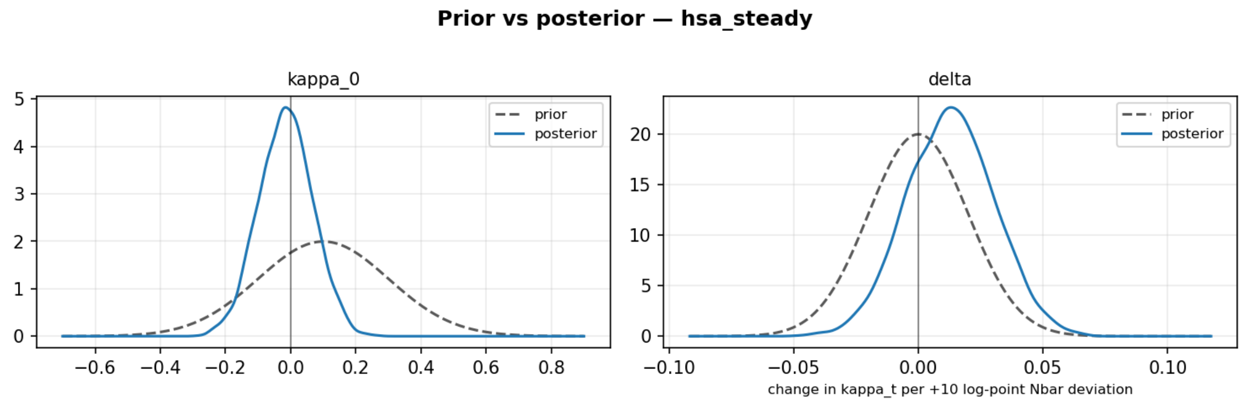
parameter	ces	hsa_steady	hsa_dynamic	hsa_full
alpha	0.640 [0.536, 0.747]	0.636 [0.529, 0.742]	0.641 [0.540, 0.744]	0.640 [0.533, 0.748]
kappa	-0.008 [-0.168, 0.150]	NaN	-0.026 [-0.181, 0.139]	NaN
kappa_0	NaN	-0.015 [-0.177, 0.145]	NaN	-0.009 [-0.179, 0.150]
delta	NaN	0.013 [-0.021, 0.047]	NaN	0.014 [-0.022, 0.048]
theta	NaN	NaN	0.035 [-0.247, 0.357]	NaN
theta_0	NaN	NaN	NaN	0.050 [-0.161, 0.291]
gamma	NaN	NaN	NaN	-0.001 [-0.038, 0.037]
rho_1	NaN	0.267 [0.056, 0.548]	0.407 [0.080, 1.117]	0.297 [0.055, 0.612]
rho_2	NaN	-0.831 [-0.969, -0.268]	-0.614 [-0.952, 0.060]	-0.833 [-0.967, -0.256]
phi_1	0.632 [0.498, 0.768]	0.633 [0.504, 0.772]	0.633 [0.503, 0.760]	0.633 [0.496, 0.764]
lambda_ez	0.010 [-0.202, 0.218]	0.019 [-0.187, 0.228]	NaN	0.017 [-0.188, 0.225]
n	NaN	-0.033 [-0.062, 0.002]	-0.030 [-0.066, 0.007]	-0.031 [-0.060, 0.002]

7.2 Prior vs Posterior by Model

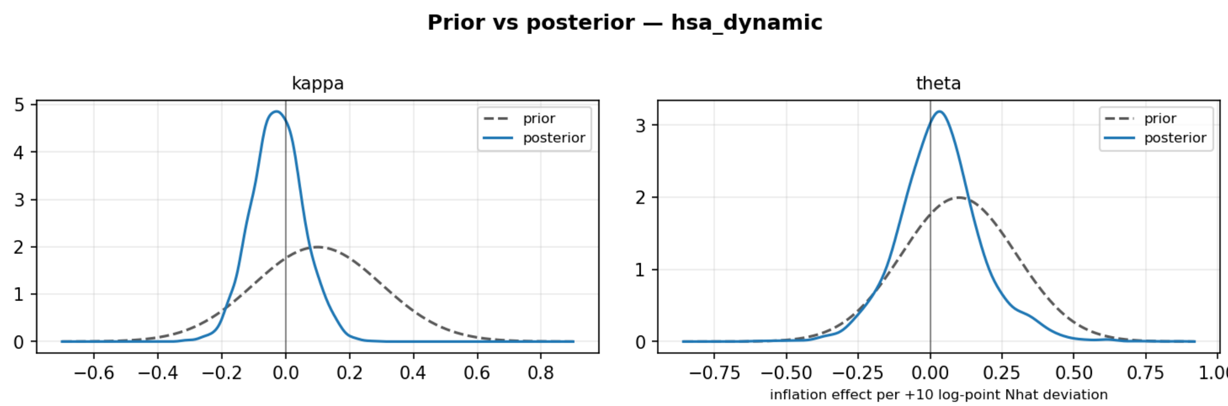
ces



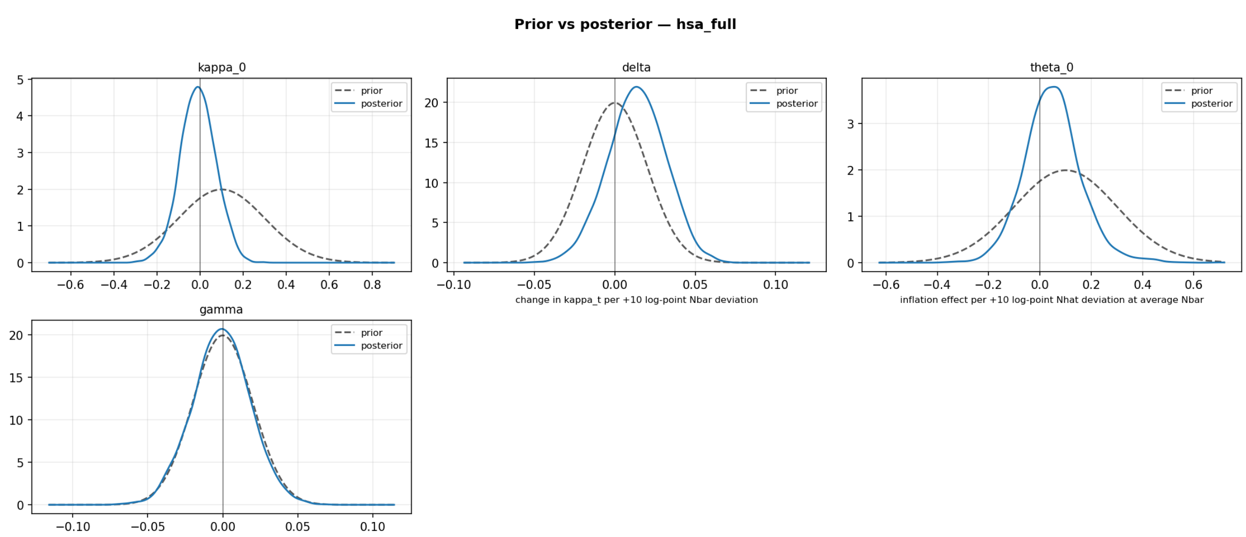
hsa_steady



hsa_dynamic



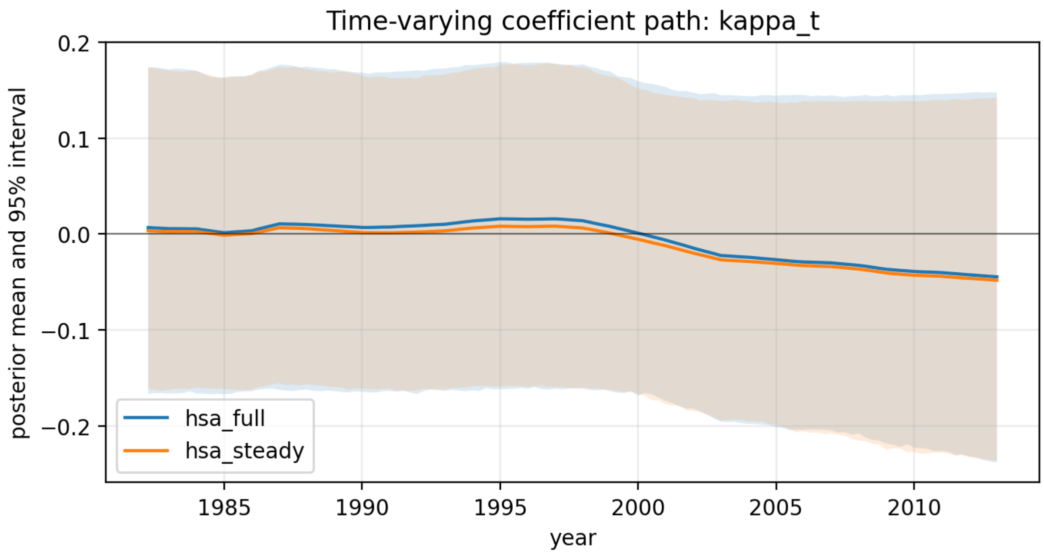
hsa_full

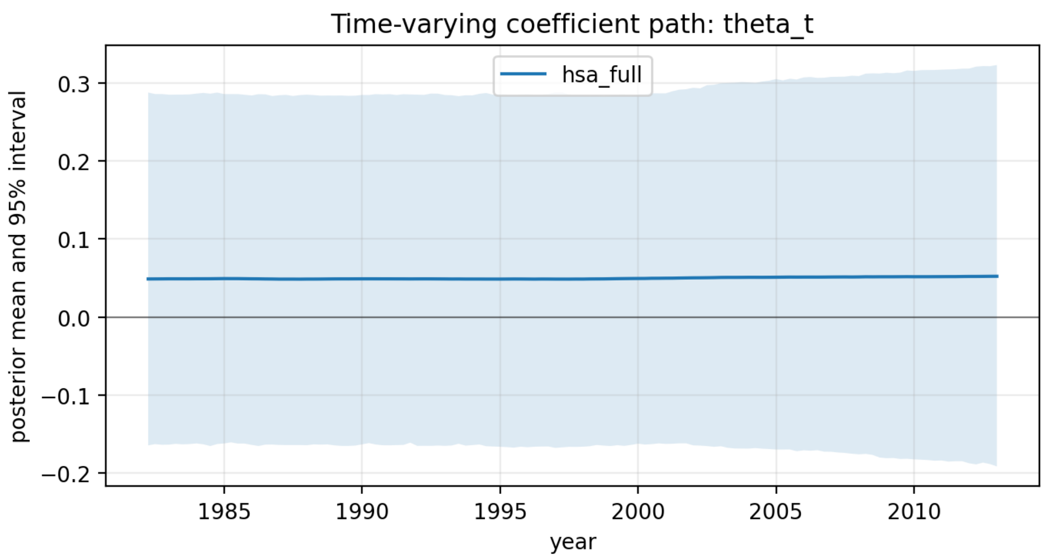


7.3 Savage-Dickey Density Ratio

model	restriction	prior_mean	prior_sd	sddr_bf01
ces	kappa=0	0.1000	0.2000	2.7022
hsa_dynamic	kappa=0	0.1000	0.2000	2.6512
hsa_dynamic	theta=0	0.1000	0.2000	1.7171
hsa_full	kappa_0=0	0.1000	0.2000	2.6975
hsa_full	delta=0	0.0000	0.0200	0.7993
hsa_full	theta_0=0	0.1000	0.2000	1.9946
hsa_full	gamma=0	0.0000	0.0200	1.0376
hsa_steady	kappa_0=0	0.1000	0.2000	2.6939
hsa_steady	delta=0	0.0000	0.0200	0.8654

7.4 Time-Varying Kappa Path





7.5 Model Comparison

model	predictive_score	posterior_predictive_rmse	log_marginal_likelihood	bayes_factor_vs_baseline	sddr_delta_bf01	sddr_theta_bf01	sddr_gamma_bf01
ces	-121.9287	0.6455	-295.1854	1.0000	NaN	NaN	NaN
hsa_dynamic	-121.9878	0.6469	773.4269	inf	NaN	1.7171	NaN
hsa_full	-121.3553	0.6422	-127.9978	4061141107206513280083379217067211658296230306556506708733926754064793600.0000	0.7993	NaN	1.0376
hsa_steady	-121.4050	0.6428	1209.2503	inf	0.8654	NaN	NaN

8 labor_share_gap_hp: quarterly_interpolated, prior=baseline, period=full, constraint=kappa ¿= 0

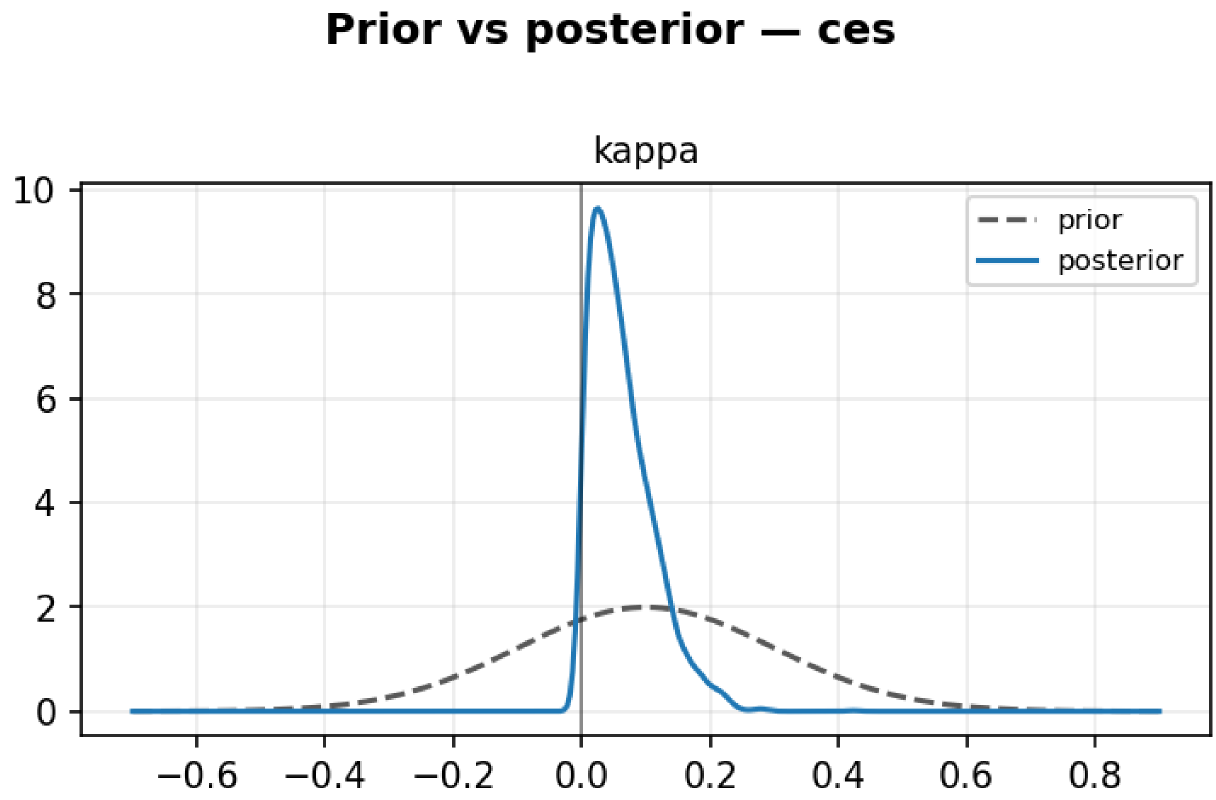
Condition. data=`labor_share_gap_hp`, competition=`quarterly_interpolated`, prior=`baseline`, period=`full`, constraint=`restricted_kappa` (kappa ¿= 0).

8.1 Coefficient Table

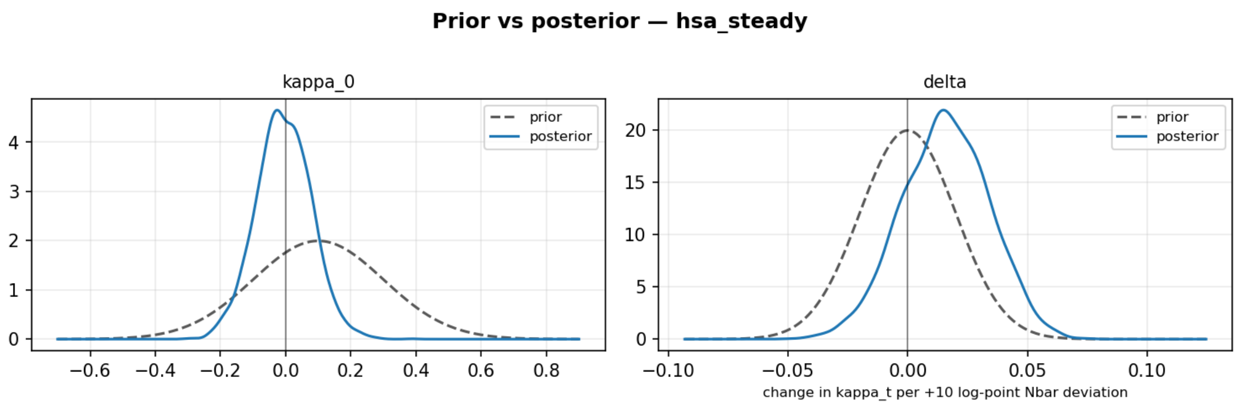
parameter	ces	hsa_steady	hsa_dynamic	hsa_full
alpha	0.629 [0.523, 0.736]	0.640 [0.537, 0.742]	0.610 [0.505, 0.715]	0.587 [0.480, 0.699]
kappa	0.063 [0.002, 0.183]	NaN	0.064 [0.002, 0.184]	NaN
kappa_0	NaN	-0.006 [-0.173, 0.157]	NaN	-0.003 [-0.165, 0.164]
delta	NaN	0.016 [-0.020, 0.050]	NaN	0.009 [-0.027, 0.044]
theta	NaN	NaN	-0.060 [-0.380, 0.541]	NaN
theta_0	NaN	NaN	NaN	0.244 [0.068, 0.477]
gamma	NaN	NaN	NaN	-0.006 [-0.045, 0.034]
rho_1	NaN	0.744 [0.370, 1.051]	0.905 [0.634, 1.121]	0.986 [0.767, 1.203]
rho_2	NaN	-0.026 [-0.309, 0.239]	0.011 [-0.195, 0.231]	-0.041 [-0.252, 0.168]
phi_1	0.632 [0.498, 0.763]	0.632 [0.499, 0.767]	0.630 [0.503, 0.763]	0.632 [0.494, 0.765]
lambda_ez	-0.059 [-0.229, 0.098]	0.017 [-0.193, 0.224]	NaN	0.018 [-0.181, 0.215]
n	NaN	-0.022 [-0.084, 0.038]	-0.036 [-0.082, 0.019]	-0.001 [-0.053, 0.051]

8.2 Prior vs Posterior by Model

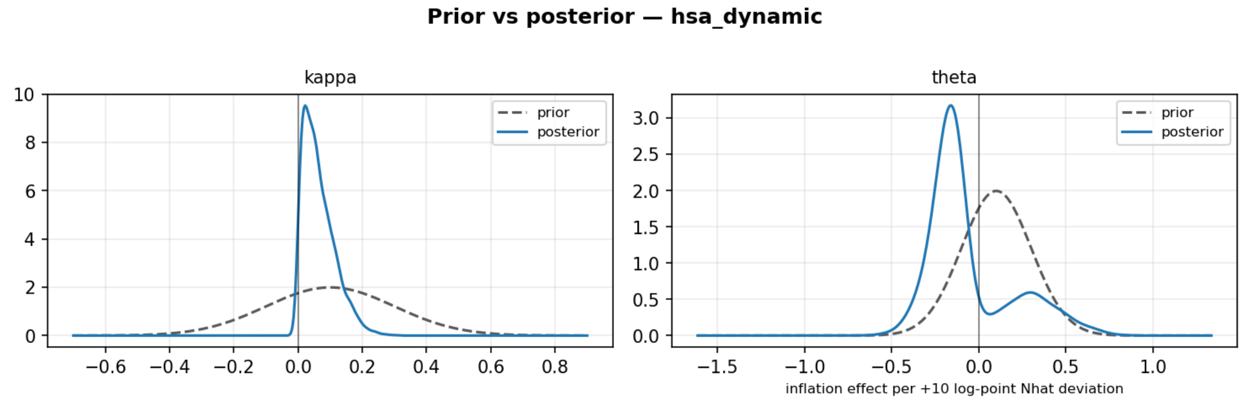
ces



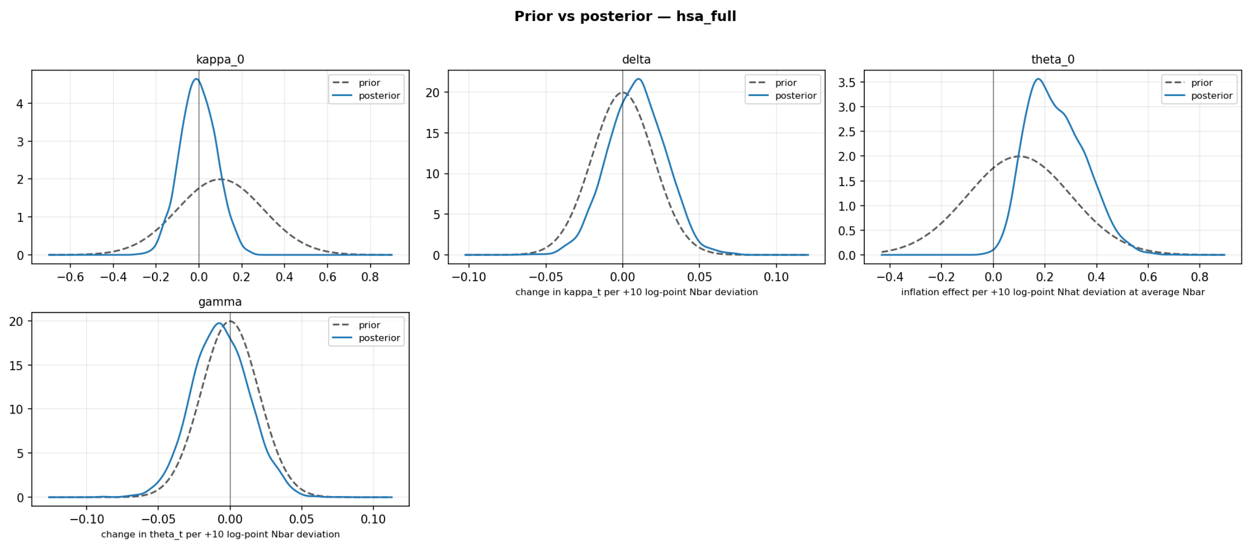
hsa_steady



hsa_dynamic



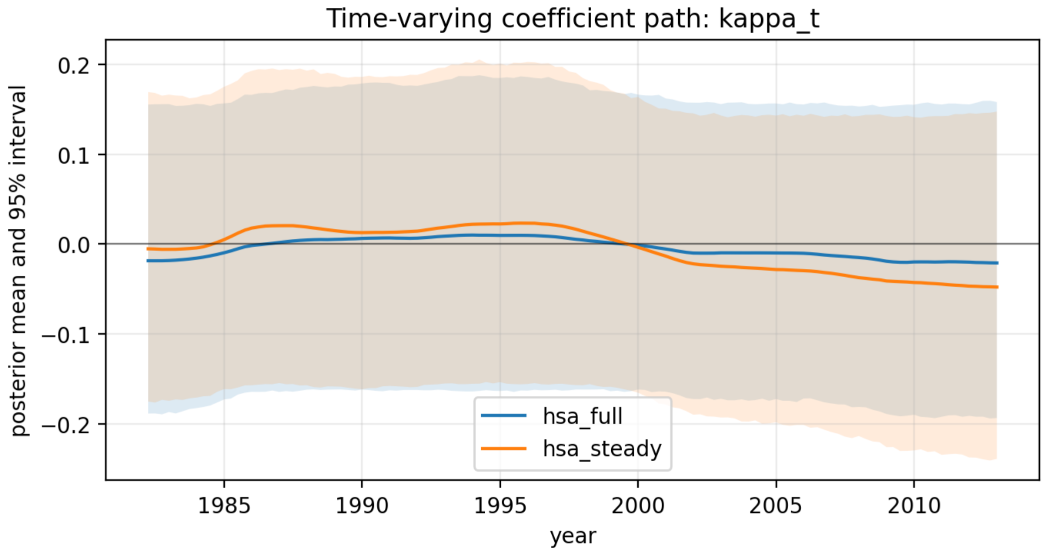
hsa_full

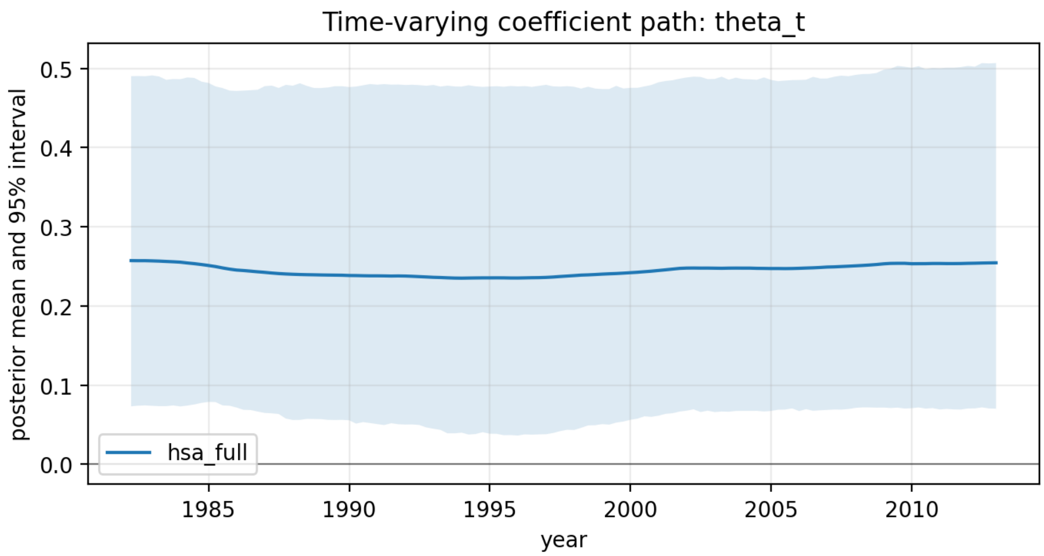


8.3 Savage-Dickey Density Ratio

model	restriction	prior_mean	prior_sd	sddr_bf01
ces	kappa=0	0.1000	0.2000	2.8874
hsa_dynamic	kappa=0	0.1000	0.2000	2.7933
hsa_dynamic	theta=0	0.1000	0.2000	0.3011
hsa_full	kappa_0=0	0.1000	0.2000	2.6073
hsa_full	delta=0	0.0000	0.0200	0.9375
hsa_full	theta_0=0	0.1000	0.2000	0.0594
hsa_full	gamma=0	0.0000	0.0200	0.9031
hsa_steady	kappa_0=0	0.1000	0.2000	2.5211
hsa_steady	delta=0	0.0000	0.0200	0.7424

8.4 Time-Varying Kappa Path





8.5 Model Comparison

model	predictive_score	posterior_predictive_rmse	log_marginal_likelihood	bayes_factor_vs_baseline	sddr_delta_bf01	sddr_theta_bf01	sddr_gamma_bf01
ces	-122.7763	0.6503	-295.5721	1.0000	NaN	NaN	NaN
hsa_dynamic	-121.0586	0.6418	-426.5227	0.0000	NaN	0.3011	NaN
hsa_full	-111.7415	0.5927	-118.4220	861575452484389047083834400302369299725247078810182760973246391963116025937 92.0000	0.9375	NaN	0.9031
hsa_steady	-121.1657	0.6414	-404.3333	0.0000	0.7424	NaN	NaN

9 labor_share_gap_hp: quarterly_interpolated, prior=baseline, period=full, constraint=kappa t restricted

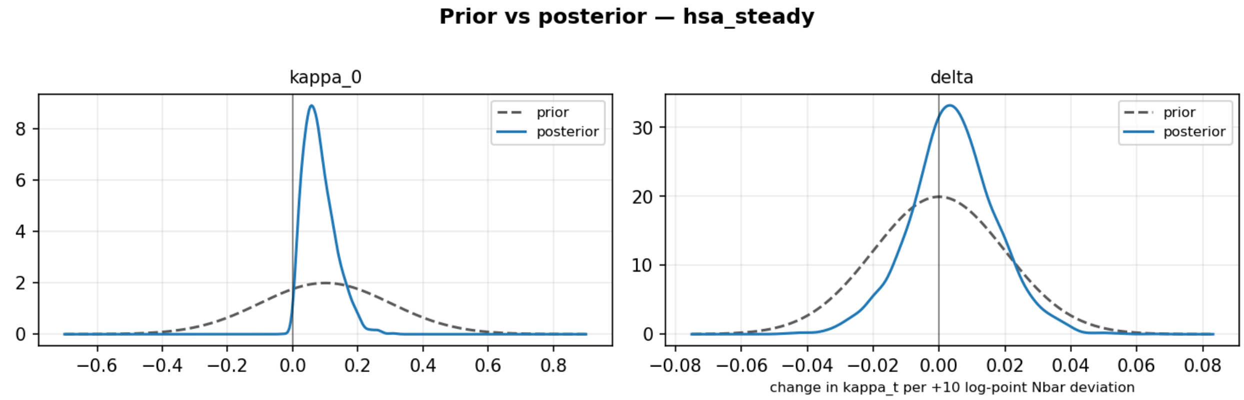
Condition. data=`labor_share_gap_hp`, competition=`quarterly_interpolated`, prior=`baseline`, period=`full`, constraint=`restricted_kappa_t` (kappa t restricted).

9.1 Coefficient Table

parameter	hsa_steady
alpha	0.627 [0.520, 0.729]
kappa_0	0.082 [0.013, 0.194]
delta	0.004 [-0.022, 0.031]
rho_1	1.807 [1.713, 1.897]
rho_2	-0.820 [-0.906, -0.727]
phi_1	0.631 [0.497, 0.768]
lambda_ez	-0.076 [-0.251, 0.085]
n	-0.044 [-0.065, -0.016]

9.2 Prior vs Posterior by Model

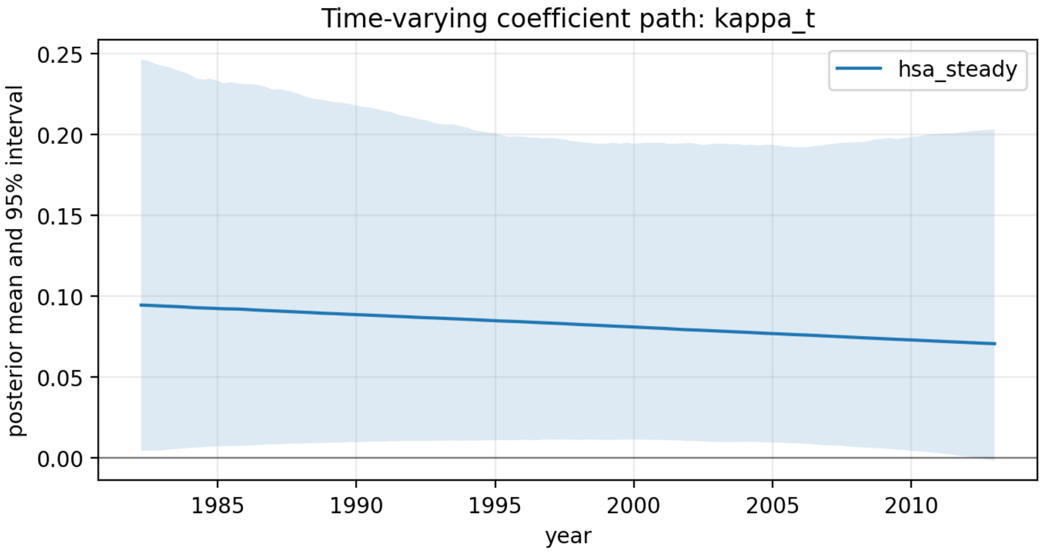
hsa_steady



9.3 Savage-Dickey Density Ratio

model	restriction	prior_mean	prior_sd	sddr_bf01
hsa_steady	kappa_0=0	0.1000	0.2000	0.6628
hsa_steady	delta=0	0.0000	0.0200	1.5853

9.4 Time-Varying Kappa Path



No theta_t path for this block.

9.5 Model Comparison

model	predictive_score	posterior_predictive_rmse	log_marginal_likelihood	bayes_factor_vs_baseline	sddr_delta_bf01	sddr_theta_bf01	sddr_gamma_bf01
hsa_steady	-123.2207	0.6524	-172.6738	1.0000	1.5853	NaN	NaN

10 labor_share_gap_hp: quarterly_interpolated, prior=weak, period=full

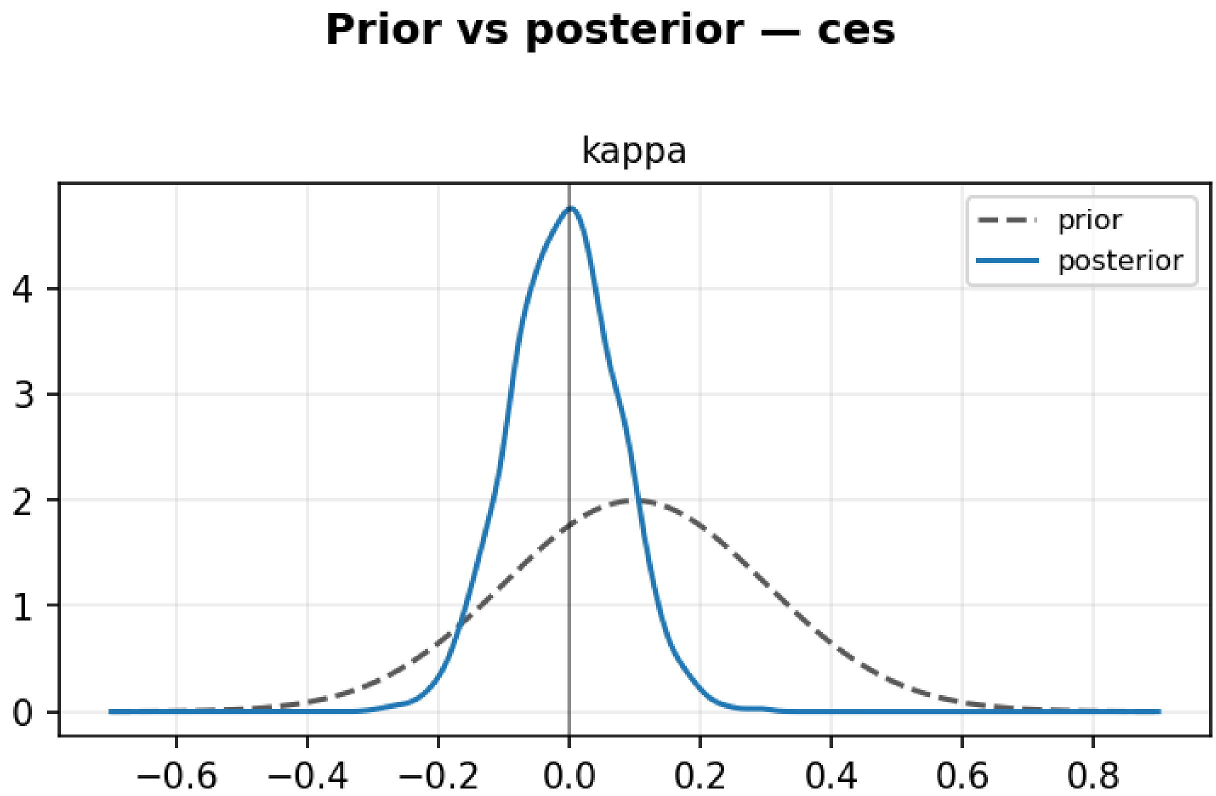
Condition. data=`labor_share_gap_hp`, competition=`quarterly_interpolated`, prior=`weak`, period=`full`, constraint=`unrestricted` (unrestricted).

10.1 Coefficient Table

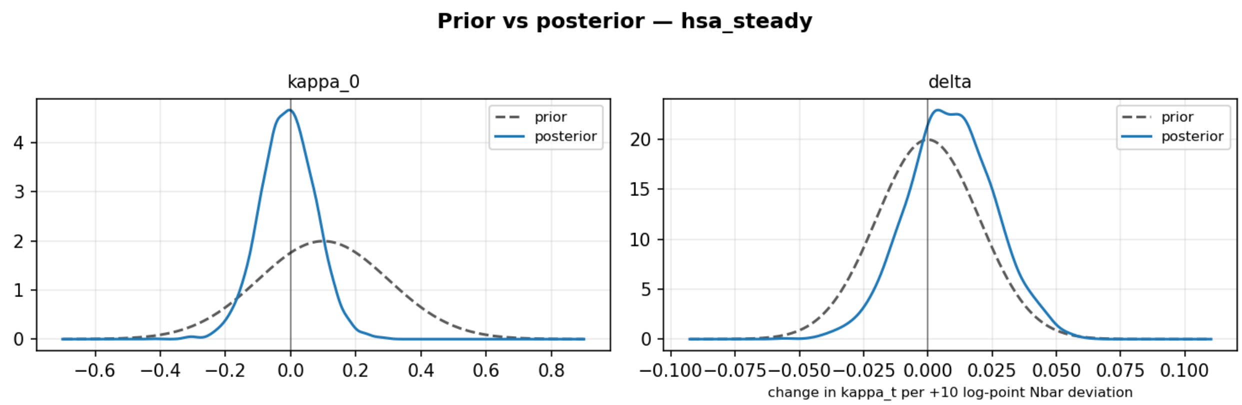
parameter	ces	hsa_steady	hsa_dynamic	hsa_full
alpha	0.640 [0.536, 0.747]	0.636 [0.530, 0.741]	0.637 [0.534, 0.737]	0.630 [0.523, 0.737]
kappa	-0.008 [-0.168, 0.150]	NaN	-0.003 [-0.164, 0.157]	NaN
kappa_0	NaN	-0.009 [-0.178, 0.157]	NaN	-0.000 [-0.160, 0.161]
delta	NaN	0.009 [-0.025, 0.042]	NaN	0.010 [-0.023, 0.043]
theta	NaN	NaN	-0.082 [-0.172, 0.026]	NaN
theta_0	NaN	NaN	NaN	-0.014 [-0.138, 0.102]
gamma	NaN	NaN	NaN	-0.027 [-0.063, 0.009]
rho_1	NaN	1.804 [1.702, 1.896]	1.767 [1.657, 1.865]	1.786 [1.672, 1.881]
rho_2	NaN	-0.817 [-0.907, -0.721]	-0.781 [-0.876, -0.676]	-0.802 [-0.894, -0.692]
phi_1	0.632 [0.498, 0.768]	0.632 [0.503, 0.768]	0.633 [0.503, 0.764]	0.632 [0.497, 0.762]
lambda_ez	0.010 [-0.202, 0.218]	0.014 [-0.199, 0.215]	NaN	0.005 [-0.200, 0.208]
n	NaN	-0.045 [-0.064, -0.021]	-0.053 [-0.071, -0.030]	-0.043 [-0.054, -0.033]

10.2 Prior vs Posterior by Model

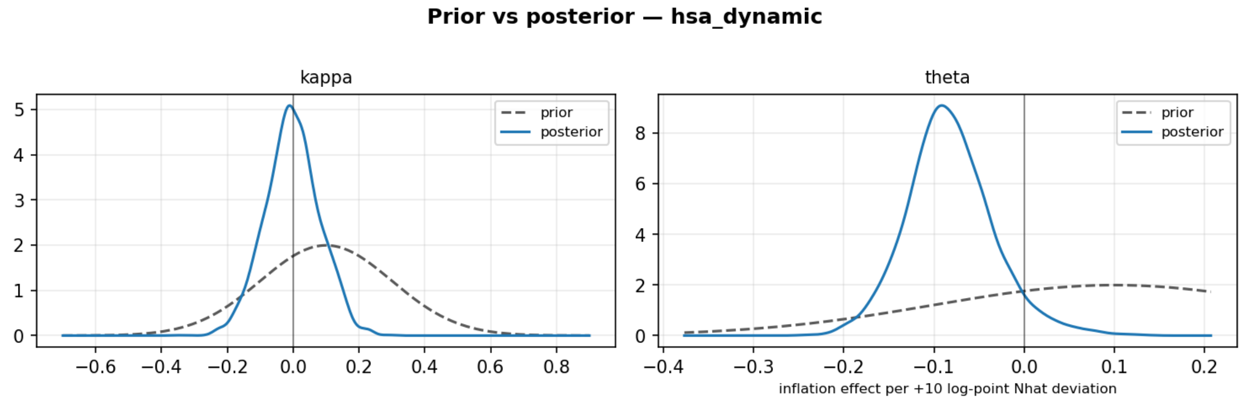
ces



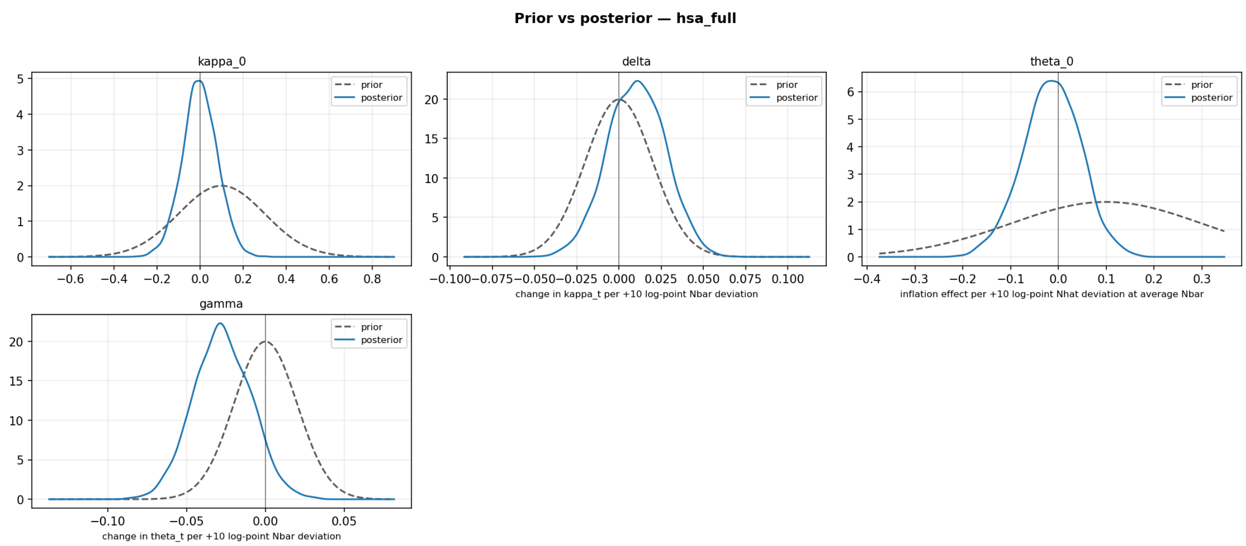
hsa_steady



hsa_dynamic



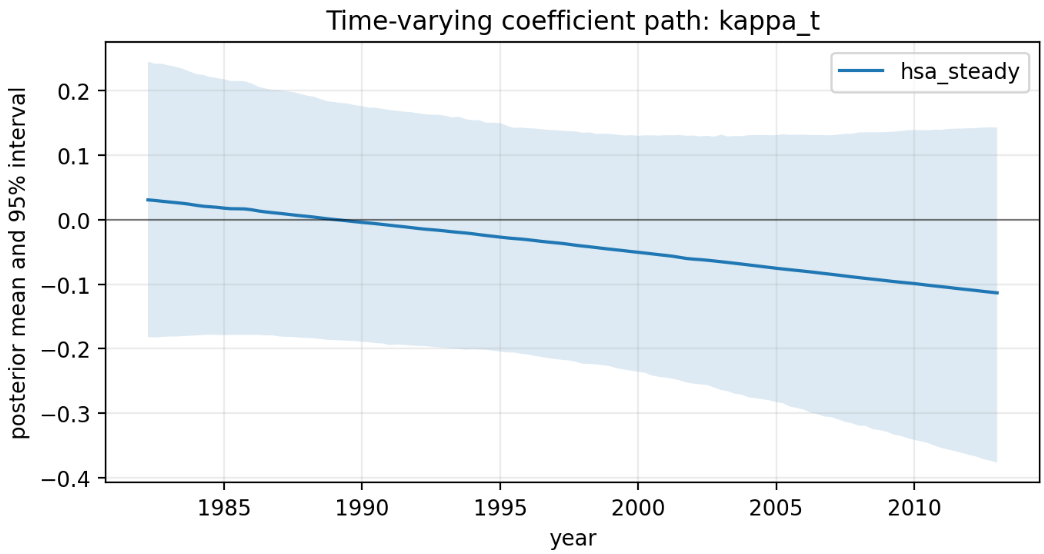
hsa_full



10.3 Savage-Dickey Density Ratio

model	restriction	prior_mean	prior_sd	sddr_bf01
hsa_steady	kappa_0=0	0.1000	0.5000	4.8110
hsa_steady	delta=0	0.0000	0.1000	1.7354

10.4 Time-Varying Kappa Path



No theta_t path for this block.

10.5 Model Comparison

model	predictive_score	posterior_predictive_rmse	log_marginal_likelihood	bayes_factor_vs_baseline	sddr_delta_bf01	sddr_theta_bf01	sddr_gamma_bf01
hsa_steady	-121.5768	0.6438	-164.8511	1.0000	1.7354	NaN	NaN

11 labor_share_gap_hp: quarterly_interpolated, prior=tight, period=full

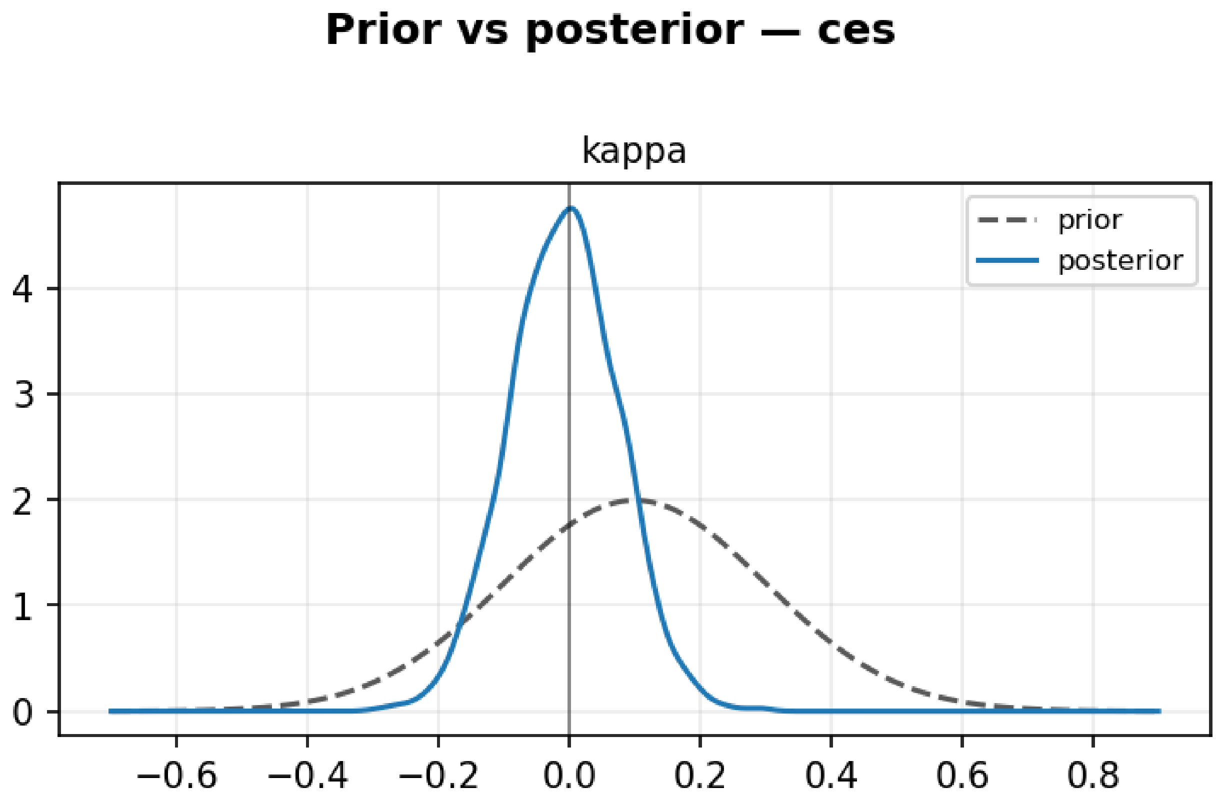
Condition. data=`labor_share_gap_hp`, competition=`quarterly_interpolated`, prior=`tight`, period=`full`, constraint=`unrestricted` (unrestricted).

11.1 Coefficient Table

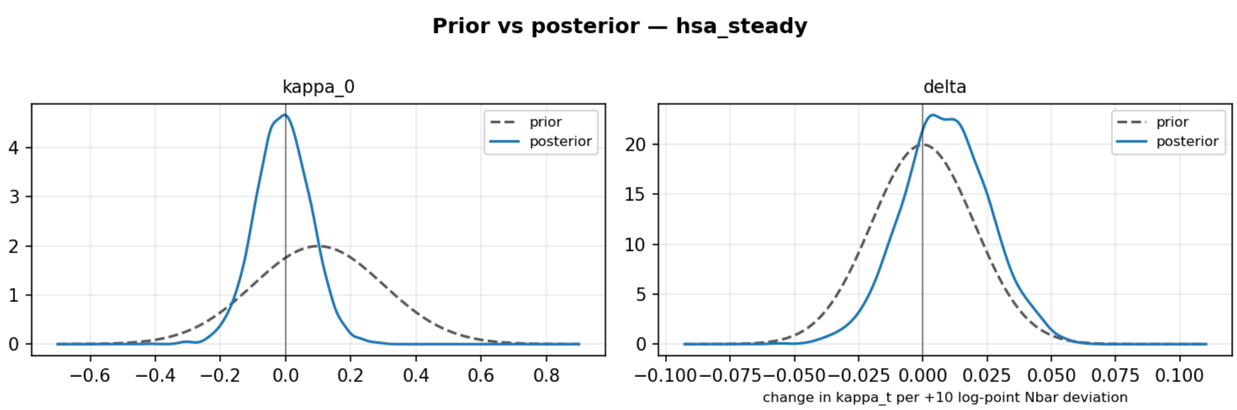
parameter	ces	hsa_steady	hsa_dynamic	hsa_full
alpha	0.640 [0.536, 0.747]	0.636 [0.530, 0.741]	0.637 [0.534, 0.737]	0.630 [0.523, 0.737]
kappa	-0.008 [-0.168, 0.150]	NaN	-0.003 [-0.164, 0.157]	NaN
kappa_0	NaN	-0.009 [-0.178, 0.157]	NaN	-0.000 [-0.160, 0.161]
delta	NaN	0.009 [-0.025, 0.042]	NaN	0.010 [-0.023, 0.043]
theta	NaN	NaN	-0.082 [-0.172, 0.026]	NaN
theta_0	NaN	NaN	NaN	-0.014 [-0.138, 0.102]
gamma	NaN	NaN	NaN	-0.027 [-0.063, 0.009]
rho_1	NaN	1.804 [1.702, 1.896]	1.767 [1.657, 1.865]	1.786 [1.672, 1.881]
rho_2	NaN	-0.817 [-0.907, -0.721]	-0.781 [-0.876, -0.676]	-0.802 [-0.894, -0.692]
phi_1	0.632 [0.498, 0.768]	0.632 [0.503, 0.768]	0.633 [0.503, 0.764]	0.632 [0.497, 0.762]
lambda_ez	0.010 [-0.202, 0.218]	0.014 [-0.199, 0.215]	NaN	0.005 [-0.200, 0.208]
n	NaN	-0.045 [-0.064, -0.021]	-0.053 [-0.071, -0.030]	-0.043 [-0.054, -0.033]

11.2 Prior vs Posterior by Model

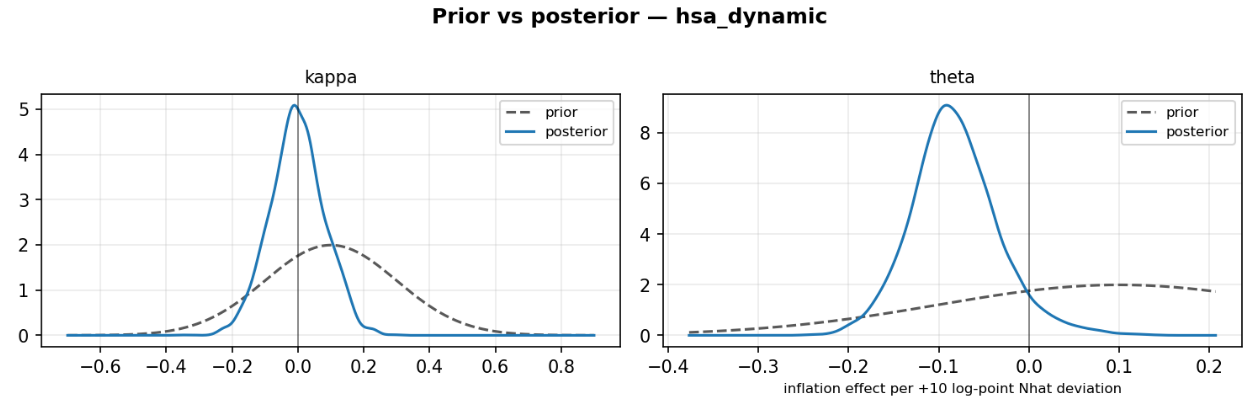
ces



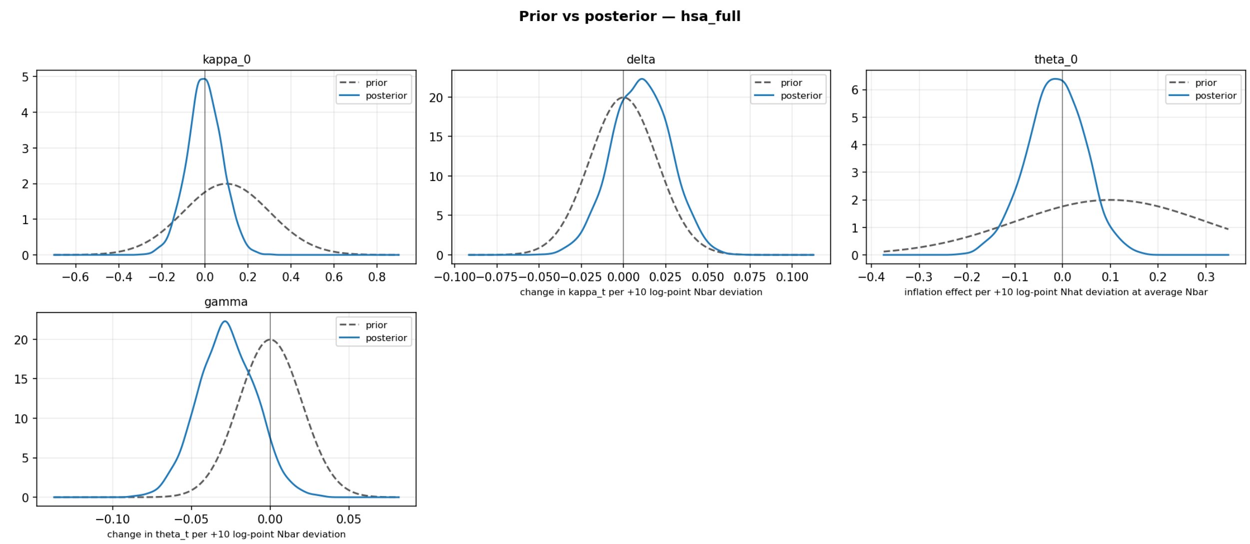
hsa_steady



hsa_dynamic



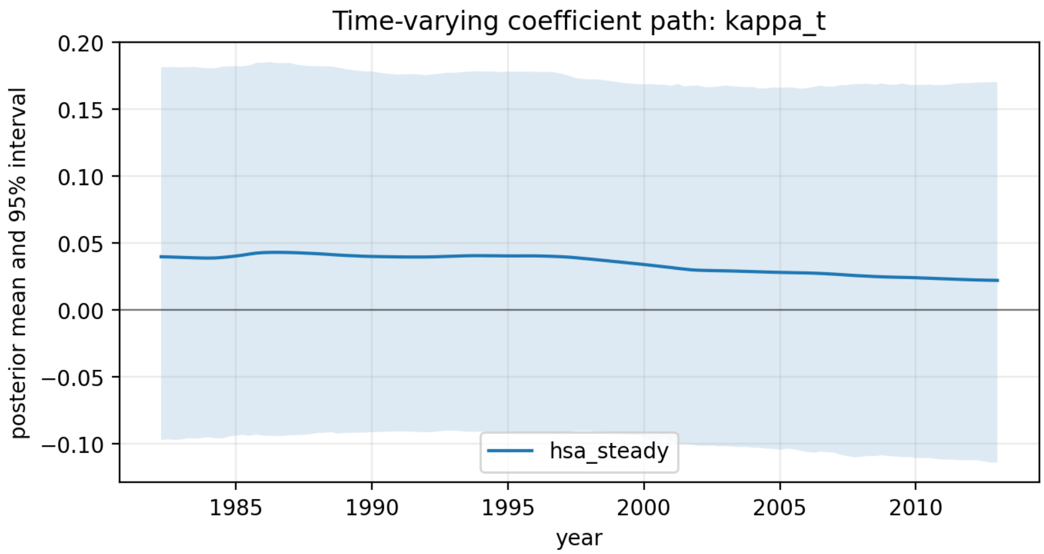
hsa_full



11.3 Savage-Dickey Density Ratio

model	restriction	prior_mean	prior_sd	sddr_bf01
hsa_steady	kappa_0=0	0.1000	0.1000	2.2291
hsa_steady	delta=0	0.0000	0.0100	0.9725

11.4 Time-Varying Kappa Path



No theta_t path for this block.

11.5 Model Comparison

model	predictive_score	posterior_predictive_rmse	log_marginal_likelihood	bayes_factor_vs_baseline	sddr_delta_bf01	sddr_theta_bf01	sddr_gamma_bf01
hsa_steady	-122.4170	0.6470	-244.9349	1.0000	0.9725	NaN	NaN

12 Competition Decomposition

For annual-Q4 mixed-frequency HSA runs, Table 2 reports selected quarters of the posterior decomposition $N_t = \bar{N}_t + \hat{N}_t$. The full quarterly decomposition is written to the corresponding CSV artifact.

Table 2: Posterior decomposition of competition N_t

quarter	model	N_total_mean	N_total_q05	N_total_q50	N_total_q95	Nbar_mean	Nhat_mean
1982Q1	hsa_full	3.2267	-1.3272	3.3666	6.5856	1.2840	1.9427
1984Q1	hsa_full	1.0175	-0.4673	1.0471	2.5003	1.0722	-0.0548
1986Q1	hsa_full	0.1825	-0.7391	0.1401	1.2073	1.0948	-0.9123
1988Q2	hsa_full	0.9293	-0.1527	0.8678	2.0617	1.3599	-0.4306
1990Q2	hsa_full	1.1735	0.4290	1.1851	1.8973	1.2025	-0.0290
1992Q2	hsa_full	1.5384	0.5435	1.5680	2.4486	1.3737	0.1647
1994Q2	hsa_full	1.7053	0.1423	1.8247	2.4747	1.7476	-0.0424
1996Q2	hsa_full	1.5400	-0.3109	1.6991	2.3191	1.8048	-0.2648
1998Q3	hsa_full	1.1961	0.1440	1.2725	1.8186	1.3470	-0.1509
2000Q3	hsa_full	0.0745	-0.7877	0.1414	0.6057	0.3389	-0.2644
2002Q3	hsa_full	-0.8854	-1.4691	-0.8724	-0.3419	-0.8010	-0.0844
2004Q3	hsa_full	-1.1037	-1.6290	-1.1043	-0.5548	-1.2121	0.1084
2006Q3	hsa_full	-1.4539	-1.9462	-1.4520	-0.9721	-1.5017	0.0478
2008Q4	hsa_full	-2.0156	-2.1231	-2.0177	-1.8968	-2.0030	-0.0126
2010Q4	hsa_full	-2.3051	-2.4231	-2.3039	-2.1924	-2.2637	-0.0415
2012Q4	hsa_full	-2.6264	-2.7462	-2.6263	-2.5115	-2.5861	-0.0403

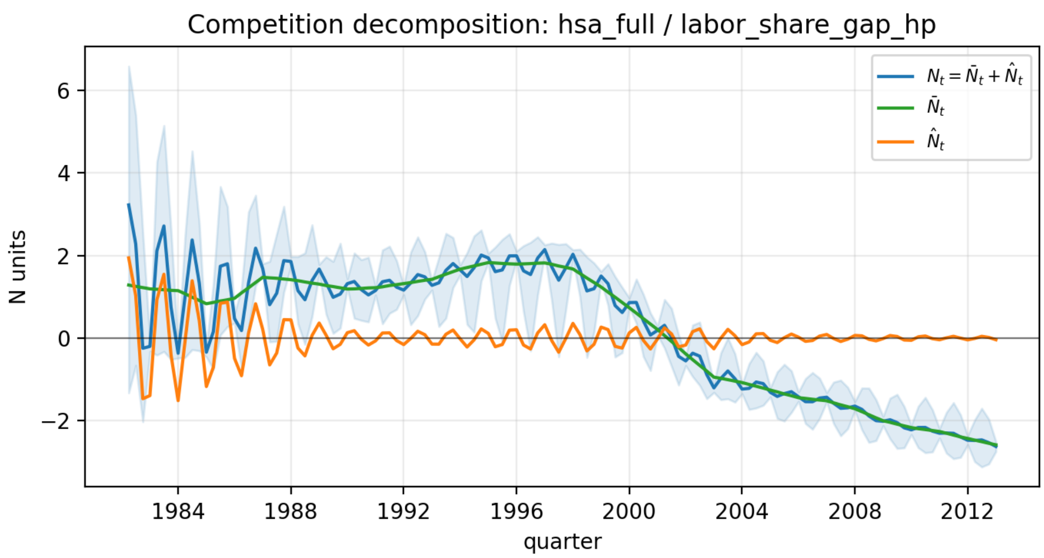


Figure 1: Posterior mean decomposition of competition

13 Notes

WAIC and LOO are not used as primary criteria because the models contain latent states. Chib marginal likelihoods are reported only when the legacy ordinate calculation can be matched to the sampler units and supplied data. For the full HSA model, the reported Chib quantity is conditional on posterior mean latent states because the observation equation is nonlinear in $\hat{N}_t$ and $\bar{N}_t$.